

Impact of Competitor Store Closures on a Major Retailer

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We use a proprietary data set from a national US department store chain to investigate the impact of competitor store closures on a major retailer. The transaction level data set includes nearly 80 million transactions, corresponding to \$2 billion in sales over a 25-month period. We find that store sales increase with respect to a store's proximity to closed competitor locations. More interestingly, we find that online channel sales also increase in geographic locations where competitors close stores and where our focal retailer has store locations in close proximity to the markets it serves. This latter finding highlights the important role that stores play in the online channel. Stores provide a level of shopping assurance generally not available online and support customer webrooming and showrooming to mitigate purchase uncertainty. Stores are also integral to omnichannel services like return-to-store wherein stores provide a convenient nearby location to make a free return should an online purchase not meet expectations. Consistent with these shopping behaviors, our results demonstrate that the focal retailer, in markets affected by competitor store closures, is able to capture demand that exhibits greater uncertainty in purchases as evidenced by (i) a disproportionate increase in riskier high value vs. low value store purchases consistent with webrooming, (ii) increased online sales of product categories associated with showrooming, and (iii) an increase in return-to-store instances for online purchases.

Key words: store closure; competition; omnichannel retailing; business analytics

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1. Introduction

Our research investigates the impact of competitor store closures on a major retailer. While market entry and store openings for retail have garnered considerable attention from both practitioners and academics, the same cannot be said for market exit and store closures. The media are replete with stories about downsizing during difficult economic times like the great recession (e.g., Kavilanz 2009) and, more recently, the pandemic (e.g., Whiteman 2021), yet managerial guidance in the literature is sparse despite downsizing being a key business strategy for enhancing market competitiveness. Moreover, store closures are significant given their potential impact on retail business functions like operations and marketing. Intrinsically, the network of operating stores determines a retailer's market reach and coverage in geographic regions as well as the efficiency and effectiveness of logistics related decisions.

Retail store closures are on the rise (Valinsky 2021). Of course, bankruptcy and a decline in financial

health are leading causes for store closures as retailers redeploy assets to remain competitive. So too are mergers and acquisitions, wherein retail chains consolidate their networks of stores and transform themselves into larger, yet more efficient competitors. In fact, there are many reasons that contribute to the rise in observed store closures. Paving the way is the phenomenal growth in online shopping that was taking place well before the pandemic first hit in 2020. Yet, the advent of the pandemic, which effectively made in-person, store shopping a health safety concern, has only exacerbated the increasing trend of store closures and sped up the shift to online shopping (Valinsky 2021).

The effects of store closures, however, are poorly understood and complicated by the multiple purposes stores serve for consumers and retailers. For consumers, stores provide a convenient location for order fulfillment. While this is rather fundamental to offline shopping, it also holds true for online shopping. Omnichannel services like buy-online, pick-up-in-store (BOPS) and ship-to-store (STS), by their very

nature, provide consumers with fast, free, and secure delivery of online purchases. Stores also facilitate shopping behaviors like research online, purchase offline as exemplified by the common practice of webrooming (Bell et al. 2014).

Moreover, stores, through their sheer physical presence in the marketplace, provide online customers with shopping assurance. One of the key disadvantages of online shopping is the risk associated with purchase uncertainty. Uncertainty is exacerbated when purchasing online, particularly for experience goods, given a customer's inability to physically evaluate the product prior to purchase. Stores reduce uncertainty by enabling the touch and feel experience in their showrooms. Indeed, *showrooming* is a common shopping behavior whereby online shoppers first visit stores to try out products and then go online to make their purchases; clearly a practice that adds insult to injury for stores. Shopping assurance is further enhanced by the omnichannel practice of return-to-store (RTS) that allows online shoppers to return their purchases to a nearby store. If for some reason the purchase fails to meet expectations, customers can immediately make a hassle-free return (Nault and Rahman 2020).

Collectively, fast order fulfillment, free and secure delivery, plus shopping assurance, are compelling reasons that drive online customers into stores. Naturally, retailers recognize the value online customers place on stores. Indeed, one of the key ways traditional retailers compete with online competitors is by adopting omnichannel retailing services such as BOPS and RTS. These services draw customers into stores and it is that foot traffic, and the ability to convert it into sales (Perdikaki et al. 2012), that represents the lifeblood for stores (Ketzenberg and Akturk 2021). Clearly, stores play a critical role in both online and offline channels.

While research on store closures is sparse, there are a few interesting contributions to the literature that are helping to increase our understanding of how store closures affect consumer behavior and hence, impact the retailers that close them (e.g., Haans and Gijsbrechts 2010, Soysal et al. 2019, Zentner et al. 2013). We are not, however, aware of any empirical research that addresses the impact of store closures on competitors. What exactly is the competitive impact of closing stores and how does the impact differ between online and offline channels? Our research is positioned to answer these and similar questions that, in turn, will help to inform on strategies for retailers that are closing stores and for competitors responding to those closures. To do so, we use a proprietary data set from a US national department store chain (DS-A) and explore the impact of competitor store closures through a series of quasi experiments.

The transaction level data set spans 25 months and includes nearly 80 million transactions, corresponding to \$2 billion in sales. By separately analyzing DS-A's offline and online channels, we contribute to the literature in several ways.

Our results for the offline channel are consistent with prior literature showing that retailers are positioned to pick up at least some business from a competitor that is closing stores (Güler 2018, Haans and Gijsbrechts 2010). Not surprisingly, the increase in sales at DS-A is influenced by the proximity of its stores to a competitor's closed store location, with the sales increase decreasing with respect to distance. Digging deeper, we find that store proximity has implications for customers who prefer to shop online and purchase in stores, through webrooming or using omnichannel services like BOPS. When stores close, customers who engage in these shopping habits are more likely to switch to a nearby competitor and our results are consistent with this explanation. Our results are also consistent with the value that online customers place on the shopping assurance that stores provide. Price is a good proxy for purchase risk as evidenced by the mounting evidence in the literature showing that higher priced items are returned more frequently than lower priced items (Akturk et al. 2018, De et al. 2013, Hess and Mayhew 1997). We observe that the sales increase at DS-A stores is disproportionately higher for higher priced items than it is for lower priced items. In addition, there is also an increase in RTS incidents from DS-A's online channel. Together, these observations suggest that it is not just new store shoppers that DS-A is able to attract from competitors that close stores. DS-A is also able to capture new customers who (a) prefer to shop online and purchase in stores and (b) prefer to both shop and purchase online.

The finding that RTS incidents increase at DS-A stores (in close proximity to competitor closed store locations) anticipates our analysis of the online channel that demonstrates sales in that channel also increase in geographic locations where competitors close stores and where DS-A has store locations in close proximity to the markets it serves. The observation that online sales increase is linked to the proximity of stores to customers highlights the importance of online shopping to store closure decisions. Clearly, only stores that are close to customers can be relevant to online shopping. Our results with respect to RTS incidents are also consistent with this explanation, plus we find support for the relevance of showrooming. Indeed, we find that online sales increase in product categories known for showrooming, but not in product categories in which showrooming is neither meaningful nor practical.

Our results for the online and offline channels provide not only a compelling narrative for assessing the competitive effects of store closures, but also insight into the mechanics underlying those effects. These insights can help to inform managerial decision-making with respect to both the store-closure decision and the competitive reactions to store closures that we discuss in the conclusion. The rest of this study is organized as follows. In section 2, we review literature. In section 3, we present our research hypotheses. In section 4, we introduce our data, define the models, and present our findings. We present the results of robustness checks in section 5, and discuss managerial insights and suggest directions for future research in section 6.

2. Literature Review

Our research is most closely related to the literature that addresses the competitive impact of store openings and closures in retail. As for openings, the literature is pretty clear that the heightened competition that ensues has a pronounced and deleterious effect on sales of incumbent retailers. Jia (2008), for example, shows market entry by Walmart or Kmart to render 48%–58% of existing supermarkets unprofitable, and Singh et al. (2006) show incumbent retailers to lose around \$0.25 million in monthly revenue following Walmart's entry in a market. Similarly, Hwang and Park (2016) show that competing grocery stores experience around a 20% loss in weekly revenue when Walmart opens a supercenter. These contributions provide strong, consistent evidence that market entry intensifies competition and increases customer defections from incumbent retailers.

Although the sales effect may be clear, the competitive impact of market entry on metrics like product variety and service is less well understood. Regarding product variety, whereas Cachon et al. (2008) and Watson (2009) find increased competition to result in less product variety, Ren et al. (2011) show product variety to increase in the presence of nearby competitors, although it tends to decrease and become more differentiated among rivals that are collocated (within a mile of one another). As for service, there are a few contributions that empirically demonstrate an increase in service due to increased competition (Matsa 2011, Olivares and Cachon 2009), while others demonstrate a decrease in service (Amihud and Medenelson 1989, Cohen and Mazzeo 2004). Clearly, the results are mixed and this ambiguity in findings is also noted by Olivares and Cachon (2009) and Ren et al. (2011).

In our research, we address the opposite, by assessing the impact of store closures on competitors. While store openings indicate a heightened level of

competition, store closures indicate the reverse. One might therefore be led to believe that the effect of a store closure will also be the reverse of an opening. Yet, there are notable differences, key among them is that with a closure, customers are forced to find a new alternative, while an opening presents a new, *optional* alternative. Hence, it is unclear how store closures and openings may differentially affect customer shopping habits, and vice-versa. The literature is largely silent on these issues, particularly with regard to the effect on competitors, to which our research is positioned.

There are a few empirical contributions that do address the impact of a retailer's store closures on its own operations. Haans and Gijsbrechts (2010) study the impact of store closures on the overall sales of a grocery retailer and find that losses arising from a store closure across different outlets depend on the store type, distance to competitors, customer profile, and purpose of the shopping trip. Güler (2018) studies the impact of a firm's mass store closures on its existing store locations. Using Starbucks mass store closure data set and running counterfactual analyses, the author estimates that the average rate of cannibalization due to store density for an average store is 4.8%, which ranges between 0% and 50% across different locations. The higher the cannibalization, the smaller the loss due to store closures. Further exploring customer churn, Soysal et al. (2019) employ a transaction level data set from a video rental firm and show that store closure increases the churn rate around 50% for users that frequently use the physical store while it does not have any effect on light users. In another study of a video rental firm, Zentner et al. (2013) study the relationship between channel choice and product choice of consumers. The authors show that when consumers move to the online channel due to store closures, they are more prone to order niche movies relative to hits due to direct channel influence. Although these studies provide valuable insights regarding the impact of store closures on a focal firm's operations, they mostly ignore the effect on competing retailers.

There is also a healthy stream of literature in the analytical, math modeling domain, that addresses the competitive facility location problem. This literature dates back to Hotelling (1929) and predominantly addresses the location of new facilities or the relocation of existing ones. Comprehensive reviews can be found in Berman et al. (2009) and Drezner (2014). More recently, researchers have begun to introduce models that optimize store closure and related decisions. For example, defining store liquidation as a time-constrained divestment, Craig and Raman (2016) introduces a stochastic dynamic program that optimizes key store liquidation decisions such as price

promotions, transfer of inventory, and timing of store closures. The authors also introduce a practical, well-performing heuristic and apply their methods within a set of case studies.

Although Craig and Raman (2016) do not address competitive effects, others do in what have become known as competitive delocation models. The premise for these models is that because retail stores operate in a competitive environment, store closure decisions are a function not only of a retailer's existing network of stores, but also of the location of stores operated by competitors. For example, ReVelle et al. (2007) introduce a mixed-integer programming model that optimizes the closing of facilities in a region so as to minimize market share lost to competitors, and Ruiz-Hernández et al. (2017), pursuing a similar objective and observing similar constraints, employ a game theoretic approach modeling Cournot and Stackelberg games for competitive delocation. In another game theoretic model, Mayadunne et al. (2018) examine the competitive downsizing decisions of two retailers and find that the firm with larger sales will push its competitor to close a greater number of unprofitable stores.

Our research complements these competitive facility location and delocation models. Our empirical results demonstrate that customer shopping behaviors are integral to both store and channel selection. For example, our finding that a retailer is likely to lose online sales when closing stores suggests that these models may be missing or miss-estimating key competitive and behavioral effects. This highlights one of our key contributions: informing the design and implementation of decision support models for store closures. We discuss these implications and associated managerial insights in the conclusion.

3. Hypotheses Development

In this section, we develop a set of hypotheses to direct our research. To do so, we draw upon the theory of transaction cost economics (transaction cost theory), as has been done in other empirical work on retail operations (e.g., Akturk et al. 2018, Grønhaug and Gilly 1991, Griffis et al. 2012) to explain consumer shopping behavior. Based on the principles of bounded rationality and opportunism, transaction cost theory posits that buyers and sellers will attempt to reduce transaction costs to their benefit (Williamson 1975, 1985). A wide variety of financial and non-financial costs, beyond price, contribute to total transaction cost. These include price-type costs (delivery charges, taxes, etc.), time-type costs (search time, delivery time, etc.), and psychological-type costs (inconvenience, anxiety, etc.) (Chircu and Mahajan 2006). The sum of these individual type costs

determines the willingness of economic actors to execute commercial transactions. In the context of store closures, we proceed to use the lens of transaction cost theory to develop our hypotheses.

A store closure will require former store patrons to choose a replacement. In making that next best selection among the remaining alternatives, customers will need to balance the aforementioned transaction costs with substitution costs (Haans and Gijsbrechts 2010). Substitution costs correspond to any inferior service or product quality provided at a shopping alternative, such as a more limited product assortment. Among the factors influencing store selection, Hortman et al. (1990) find that travel distance has the greatest influence; customers tend to maximize utility by minimizing travel time and cost which contribute to overall transaction cost. As demonstrated by Haans and Gijsbrechts (2010), a retailer that closes stores may attract some of their customers to other nearby locations, but it is also likely that nearby competing retailers will be able to capture some of those customers and we refer to this as the proximity effect. Given the importance of distance, we expect the proximity effect to diminish as the distance to a closed store location increases. This reasoning, which dates to Hotelling (1929), is elegantly, if not famously, encapsulated in the Hotelling line. Hence, we hypothesize:

HYPOTHESIS 1A (H1A). *Sales will increase at a store that is in close proximity to a competitor's closed store location and the observed increase will diminish as the distance between the two increases.*

As indicated by the contributions of Soysal et al. (2019) and Zentner et al. (2013), former customers of a closed store may, instead of switching to a different store location, switch their shopping to another channel. After all, customer choice involves maximizing utility and given heterogeneous customer valuations, once a store closes the next best option may be online. This line of reasoning is a prelude to the development of a set of hypotheses on the impact of store closures on a competitor's online channel. In highlighting the value that customers place on making their purchases in stores, it also helps to explain why the Internet may fall short in delivering the desired value.

Generally, online shopping offers several advantages to consumers that include shopping convenience, lower product prices, greater variety, and a wealth of information (e.g., product descriptions, reviews, price comparisons) that facilitates product search (Rigby 2011). Yet, there are also disadvantages and a notable one is a heightened level of purchase uncertainty due to the consumers inability to touch and feel the product prior to purchase. Greater uncertainty translates to a higher likelihood of a product

mismatch online than in stores (Ofek et al. 2011). The added uncertainty is associated with risks that include product performance risk, financial risk, as well as, time and inconvenience risk (Danaher et al. 2003, Griffis et al. 2012, Peck and Childers 2003). Product performance risk relates to product quality not meeting expectations, financial risk to concerns and added expenses associated with returns (Hassan et al. 2006), and time and inconvenience risk to the delivery process (Forsythe et al. 2006). Collectively, these risks drive transaction costs.

A lower likelihood of purchasing online, whether due to the inherent risks, or perhaps some other disadvantages associated with online shopping, does not preclude consumers from searching online to take advantage of the information available on the Internet and then later making their purchase in a store. Such research online, purchase offline shopping behaviors are known as webrooming and according to a Harris poll, 69% of consumers engage in webrooming (Khan 2018). Given the relative advantages and disadvantages of shopping in offline and online channels, certain products and product categories are more or less prone to webrooming than others. Low priced items for which there is little purchase risk or similarly, everyday consumables and repeat purchases that have little associated uncertainty, are examples of products less prone to webrooming. Conversely, higher priced items in which there is more risk are more prone to webrooming (Khan 2018, Manss et al. 2020). Because a store closure reduces choice for consumers who webroom and thereby increases their perceived transaction costs, we expect that a retailer will benefit more from a competitor store closure in product categories that are more prone to webrooming. Using price as an indicator of webrooming, we hypothesize:

HYPOTHESIS 1B (H1B). *The increase in store sales observed due to a competitor store closure will be larger for higher than for lower value purchases.*

Regardless of the competitive impact store closures may have on the offline channel, we expect there to be a separate and distinct impact on the online channel. Online customers who utilize omnichannel services like BOPS, STS, and RTS are likely to be negatively affected. These services, as well as other consumer shopping behaviors like showrooming, reduce the risks and costs associated with online shopping, but by their very nature, require a physical store presence in the trading area. Hence, when confronted with a store closure, customers who use these services will need to find an alternative store location, much the same as for store patrons that are affected by a store closure.

Consider the BOPS and STS omnichannel services. For online shoppers, home delivery can present difficulties that are effectively mitigated by these services since customers will instead go to a store to pick up their purchases. Shipping cost is a key disadvantage to home delivery, prior research showing that shipping fees and handling surcharges undermine the purchase intentions of online customers (Gümüş et al. 2013, Lewis et al. 2006). A recent survey indicates that shipping fees are ranked among the primary reasons for online cart abandonment (Baymard Institute 2017). Secure home delivery is another potential issue that can drive up transaction costs. Reolink (2017) reports that 23 million Americans' packages are stolen each year. In fact, there are literally countless delivery locations throughout the United States and the world at which packages should not be left unattended, high crime areas and inner cities being particularly problematic. Receiving unattended packages in major metropolitan areas can be equally vexing, and waiting at home for delivery not only inconvenient, but often infeasible. Hence, services like BOPS and STS reduce transaction costs for customers with home delivery issues.

Services like BOPS and STS also facilitate RTS that allows customers to make returns at a store. As previously discussed, a major disadvantage of online shopping is heightened purchase uncertainty and the associated risks. Customers who pick up in stores can immediately inspect the product, compensating for online shoppings lack of the touch and feel experience, and if dissatisfied make an easy and costless return. In effect, omnichannel services like BOPS enhance the perceived leniency of return policies by relieving customers of the hassles of locating the receipt, repackaging the product, creating a return label, and shipping to the online center or dropping at the nearest store. Those hassles are eliminated for the customer. The benefits attributable to RTS still largely apply even for customers who do not use BOPS and STS. A survey by UPS (2015) finds 65% of consumers consider the ability to return an online purchase to a store a positive aspect of a hassle-free experience, and Narvar (2017) reports 47% of online customers believe it is easier to return an online purchase to a physical location. Hence, customers who shop at retailers that offer RTS should feel more comfortable about the return process, which in turn will reduce both price-type and psychological-type costs (Chircu and Mahajan 2006).

Stores can be used to mitigate uncertainty even for consumers who prefer to purchase online without using an omnichannel service. The obverse of webrooming, showrooming is a common shopping behavior whereby customers evaluate products in stores, prior to purchasing them online (Mehra et al. 2018).

According to a consumer survey, 78% of shoppers regularly showroom (Retail Perceptions 2014), and in doing so effectively mitigate the absence of touch and feel in the online channel (Bell et al. 2015). Of course, simply because consumers showroom at a given retailer does not mean that they will ultimately buy online from that same retailer, but often this is exactly what happens (Khan 2018, Richter 2013). Bell et al. (2018) find that showrooms increase not only online sales, but also overall demand across channels. Confronted with a store closure, showroaming customers, just as store patrons, must find an alternative store to serve as a showroom.

Indeed, the same theoretical reasoning for H1A that explains the behavior of store patrons faced with a store closure extends to online customers who use omnichannel services or showroom. A retailer's ability to capture online customers from a competitor depends critically on the distance between those customers and the retailer's store locations, just as the distance between a retailer's store location and a competitor's closed store location is critical to capturing offline customers. Proximity to customers is impacted by the size and concentration of the market served and number of stores in the market. From this perspective, sales should increase with respect to customer proximity. Hence, we hypothesize:

HYPOTHESIS 2A (H2A). *Online channel sales will increase in geographical regions in which competitors close stores relative to geographical regions in which there are no competitor store closures and the sales increase will be greater in markets that are in close proximity to online customers.*

Of course, H2A is an aggregate measure that obfuscates the source of the sales increase. Our theoretical reasoning is predicated on the value consumers place on omnichannel services and showroaming to reduce transaction costs. A more granular analysis of each source will enhance understanding of their relative influence. With regard to omnichannel services, our focal retailer offers only RTS during the period of our analysis. As the value customers place on the shopping assurance provided by RTS should be directly observable in the number of RTS instances, we hypothesize:

HYPOTHESIS 2B (H2B). *RTS instances will increase in geographic regions in which a competitor closes a store and the increase will be greater in markets of close proximity to online customers.*

The relative advantages and disadvantages of the offline and online channels render particular product categories more or less suited to showroaming. A

study of consumer shopping behaviors by Statista reports that the two most showroomed product categories are consumer electronics and apparel, while the least most showroomed category is jewelry and watches (Khan 2018, Manss et al. 2020). Because a store closure reduces choice for consumers who showroom, we expect a retailer will benefit more from a competitor store closure in product categories that are more suited to showroaming. We thus hypothesize:

HYPOTHESIS 2C (H2C). *In geographic regions where there is a competitor store closure, online sales in product categories suited for showroaming will increase more than those not suited for showroaming, and the increase will be greater in markets that are in close proximity to online customers.*

4. Data, Models, and Analysis

Store closures by major competitors constitute an external shock to DS-A's online and offline operations. Our setting enables us to employ the difference-in-differences (DID) methodology to estimate the impact of external shocks by establishing treatment and control groups (Donald and Lang 2007, Forman et al. 2009). The treatment group requires selecting experimental units directly affected by store closures, whereas the control group consists of experimental units unaffected by store closures. With the group assignments made, the DID methodology is able to address trends in the data that could bias the results of the analysis due to potential confounding (Athey and Imbens 2006, Avery et al. 2012). A quasi-experimental setting also addresses endogeneity concerns arising from the unobserved heterogeneity of omitted variables. We can thus estimate the impact of store closures by evaluating the change in the treatment group relative to the control group pre- and post-intervention.

We recognize that the usual organization for reporting on empirical research is to first describe all of the relevant data, then second to introduce all of the models, and third, to present the results and analysis. We, however, take a different approach here. Because we conduct separate offline and online analyses, our research involves two populations with different units of analysis. Moreover, the dependent and independent variables, as well as controls, vary across models. Hence, we feel that to discuss all the data, then all the models, and lastly all the results, would be a burden to the reader. Instead, we choose to separate our discussion of the offline and online channels in order to simplify the exposition and present a more streamlined narrative. The rest of this section is organized as follows. In section 4.1 we describe our empirical data and setting, then in section 4.2 we

discuss our analyses of the offline channel, followed by the online channel analysis in section 4.3.

4.1. Empirical Setting and Data

We employ a proprietary transactional data set from DS-A to investigate the impact of store closures by two major competitors (DS-B and DS-C) on its operations. All three competitors are large national US department store chains. The data set spans 25 months, from April 2014 to April 2016, and includes 79.2 million purchase and return transactions by 12 million unique customers across more than 300 stores and an online channel. The online channel accounts for 4.9 million transactions representing \$167.7 million (8.3%) in sales. The offline channel accounts for the balance of transactions. The data set is at the transaction line-item level and includes: transaction id, customer id, store id, sku id, transaction date and time, transaction line number, transaction type (i.e., return vs. purchase), product price, product category, store location, and customer address. In addition to the transaction data, we utilize nine other data sets and explain their use as we proceed. Our subsequent analysis includes only the 282 stores that were operational throughout the study period. We later assess the impact of DS-A's own store closures and openings on our analyses in section 5.

DS-B and DS-C operate more than 700 and 800 stores, respectively, and both operate online channels. Note that all three competitors rank in the top 10 US department stores (USA Today 2017). The product categories they sell include apparel, accessories, furniture, jewelry, cosmetics, and house goods, among others. The strong competitive relationships among these three retailers are acknowledged in their annual reports and evidenced by the fact that their stores are collocated in many shopping malls throughout the United States.

Consistent with Song et al. (2015) and Akturk et al. (2018), we set aside April 2015, during which DS-B and DS-C closed more than 50 stores, as the acclimation period and do not use it in our analyses. Thus, the period from April 2014 to March 2015 represents the pre-treatment period and the period from May 2015 to April 2016 represents the post-treatment period. Throughout the study period, DS-A does not change any shipping or return policies and does not adopt any new omnichannel services.

4.2. Offline Channel Analysis

The unit of analysis is store, aggregated by month. See section 5 for robustness to other levels of aggregation. Our analysis is predicated on appropriately assigning DS-A stores to treatment and control groups. Only a nearby store can serve as a viable

alternative to a closed store as it must be readily accessible to former patrons of that store. A DS-A store near a closed store is thus assigned to the treatment group, and otherwise to the control group. We acquired closed store locations through press releases and annual reports of DS-B and DS-C and the corresponding longitude and latitude information from AggData, which sells store location data. We use the Haversine great-circle distance between two points on a sphere to calculate the distance between each DS-A store and the nearest closed store. Our choice of 20 miles as the cutoff distance for assigning stores to experimental groups, albeit somewhat arbitrary, is enough to make a DS-A store sufficiently inconvenient to visit such that influence from the nearest closed store should be negligible. See section 5 for robustness to this modeling choice.

4.2.1. Variables and Descriptive Statistics. We use the number of purchase transactions (*STORE PURCHASES*) per store, aggregated by month, as our response variable (see section 5 for robustness to this modeling choice using the dollar value of sales). We include control variables that indicate regional economic activity in the United States. At the designated market area level (*DMA*), we use unemployment (*DMA UNEMPLOYMENT*) and total households (*TOTAL HOUSEHOLDS*), obtained from Standard Rate and Data Service database managed by Kantar Media. *DMAs*, often used to measure audiences reported by Nielsen Media Research and to price advertising, are geographic regions the populations of which are exposed to the same advertising across various media including television, radio, and print. There are 210 *DMAs* across the United States.

As additional controls, we use state-level monthly unemployment rate (*STATE UNEMPLOYMENT*) acquired from the US Bureau of Labor Statistics and average household income (*HOUSEHOLD INCOME*) acquired from the US Census Bureau. To control for seasonality, we include month–year fixed effects (*MONTH-YEAR FE*). We also employ store fixed effects (*STORE FE*) to control for the time-invariant characteristics of DS-A stores. Table 1 reports descriptive statistics (means and standard deviations) for key variables of interest, separately for each experimental group, before and after the store closures.

Before testing our hypotheses, we check the parallel trends assumption for DID as the treatment and control groups may follow different (not parallel) trends in the pre-closure period. We do so by testing the interaction term, $TREAT_j * TREND_t$, where $TREAT_j$ denotes whether store j is in the treatment group and $TREND_t$ indicates the number of months from the start of the analysis period (April 2014) to the store

Table 1 Descriptive Statistics-Offline Channel

Variables	Treatment group (25 DS-A Stores)				Control group (257 DS-A Stores)			
	Before closures		After closures		Pre-closures		Post-closures	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
STORE PURCHASES (000)	7.95	3.64	8.49	4.10	9.13	4.94	9.55	5.38
STORE SALES (000)	215.42	121.47	234.33	132.39	253.05	166.97	269.06	175.20
DMA UNEMPLOYMENT (%)	5.17	0.93	4.10	0.49	4.76	1.89	3.91	1.36
TOTAL HOUSEHOLDS (000,000)	1.57	0.77	1.61	0.79	0.91	0.77	0.98	0.86
STATE UNEMPLOYMENT (%)	6.42	0.86	5.45	0.64	5.64	1.01	5.02	0.83
HOUSEHOLD INCOME (000)	66.89	5.42	67.46	5.41	68.18	6.69	68.84	6.87

closure month (April 2015). We find that the coefficient for the interaction term, $TREAT_j * TREND_t$, for store purchases is not statistically significant ($p > 0.1$, $R^2 = 0.86$). A plot of the preintervention trends (not presented) also shows no difference in trends between experimental groups. Finding no evidence that treatment and control groups follow different trends in the pre-closure period, we conclude that the parallel trend assumption for DID is not violated.

To test H1A and investigate the impact of store closures on DS-A's offline channel sales, we first categorize DS-A stores as "high" or "low" treatment intensity based on the distance to the closest DS-A store. We adopt an intensity-based measure to assess the differential impact of treatment intensity on experimental units (see Acemoglu et al. 2004, Duflo 2001, Lins et al. 2017 for details on the treatment intensity approach). Using 10 miles as the cutoff threshold for treatment intensity, there are 11 high intensity stores ($HIGH TREAT_j$) within 10 miles, and 14 low intensity stores ($LOW TREAT_j$) between 10 and 20 miles, of a closed store location. See section 5 for model robustness with respect to other cutoff distance thresholds. Letting $CLOSURE_t$ indicates whether month t is in the post-closure period (1) or in the pre-closure period (0), we have the following econometric model to test H1A:

$$\begin{aligned} \log(STORE PURCHASES_{jt}) \\ = \mu_j + \theta_t + \delta_1 LOW TREAT_j * CLOSURE_t \\ + \delta_2 HIGH TREAT_j * CLOSURE_t \\ + \delta_s CONTROLS_{sjt} + \varepsilon_{jt} \end{aligned} \quad (1)$$

where μ_j and θ_t are fixed effects for stores and month-years, respectively, and ε_{jt} denotes the random error term. The vector $CONTROLS$ includes the full set of control variables and the subscript s identifies a specific variable within the vector. We use the fixed effects (FE) model for the estimation and log transform the dependent variable due to excessive skewness in the data. The main effects of $LOW TREAT_j$, $HIGH TREAT_j$, and $CLOSURE_t$ are omitted in Equation (1) because μ_j , $LOW TREAT_j$

and $HIGH TREAT_j$ are not separately identified due to collinearity (Imbens and Wooldridge 2008). In addition, time-invariant variables are not identified with a FE model and so too $CLOSURE_t$ when we add month-year fe. The same arguments apply to Equations (2) and (3). Furthermore, Bertrand et al. (2004) note that traditional DID models significantly underestimate the standard deviations of the coefficients. To address this concern, we conduct our analyses using heteroscedasticity robust standard errors clustered at the store level. We adopt the same approach throughout subsequent analyses for the offline channel.

We report the results for Equation (1) in Table 2 and use the estimated coefficients for the two-way interaction terms, δ_1 and δ_2 , shown in the second and third rows, to test H1A. The second column (Model-B1) reports the findings when we include all treatment and control group stores. The third column (Model-B2) reports our findings after pairing experimental units through propensity score matching to arrive at equally sized experimental groups; specifically, we select the 25 of the 257 control group stores most similar to the 25 treatment stores (see section A.1 for details). As shown in Table 2 for Model-B1, the coefficients for both $LOW TREAT_j * CLOSURE_t$ and $HIGH TREAT_j * CLOSURE_t$ are positive and

Table 2 Impact of Store Closures on Offline Channel Purchases

Variables	Store purchases			
	Model-B1		Model-B2	
$LOW TREAT * CLOSURE$	0.02*	(0.010)	0.03**	(0.013)
$HIGH TREAT * CLOSURE$	0.04*	(0.017)	0.05**	(0.018)
TIME-VARYING CONTROLS	Yes		Yes	
MONTH-YEAR FE	Yes		Yes	
STORE FE	Yes		Yes	
MATCHED SAMPLE	No		Yes	
N	6768		1200	
R^2	0.86		0.83	

Note: Standard errors (in parentheses) are heteroscedasticity robust and clustered at the store level. FE denotes fixed effects.

** $p < 0.01$, * $p < 0.05$.

statistically significant ($\delta_1 = 0.02$, $p < 0.05$ and $\delta_2 = 0.04$, $p < 0.05$). The results with propensity score matching are even stronger. Collectively, both sets of results demonstrate that store closures have a positive and significant impact on the number of purchases in DS-A's offline channel. Furthermore, the magnitude of the coefficient for $LOW\ TREAT_j * CLOSURE_t$ is lower than that for $HIGH\ TREAT_j * CLOSURE_t$, indicating that the increase in purchases diminishes as the distance between DS-A stores and competitor closed store locations increases. Hence, H1A is supported.

To test H1B and determine if the increase in offline channel sales is greater for high value than for low value purchases, we use the mean product price (\$28.30) to distinguish between low and high. For our purposes, high and low are better distinguished by the mean than the median given the preponderance of low value purchases. This categorization yields 22.4 million high value purchases and 49.7 million low value purchases. We replicate our original analysis using the econometric model in Equation (1) separately for each category and present the results in Table 3. The second and third columns (Model-1 and Model-2) report the findings for high value purchases while the fourth and fifth columns (Model-3 and Model-4) report the findings for low value purchases. For each analysis, the first columns report the results when we include all DS-A stores in the analysis, the second columns report the findings after pairing experimental units via propensity score matching.

As shown in Table 3, the coefficients for $LOW\ TREAT_j * CLOSURE_t$ and $HIGH\ TREAT_j * CLOSURE_t$ are positive and statistically significant in both Model-1 and Model-2. For low value purchases, however, the coefficients are not statistically significant except for $HIGH\ TREAT_j * CLOSURE_t$ in Model-4. Collectively, these results show that the increase in store purchases due to a competitor store closure is larger for high value than for low value purchases. Hence, H1B is supported.

4.3. Online Channel Analysis

Our analysis of the online channel, like the offline channel, is predicated on appropriately assigning experimental units to treatment and control groups. As store closures can only have an impact on customers in nearby geographic regions, consistent with Gallino et al. (2017) and Akturk et al. (2018) we use DMAs to identify geographic regions. Hence, we include DMAs in which a store closure occurs (does not occur) in the treatment (control) group. Our unit of analysis is DMA with monthly aggregation. See section 5 for robustness with respect to this modeling choice. Having identified for the offline analysis the DMA in which each DS-A store is located, we now need only determine the DMA for each DS-B and DS-C closed store location, for which we use a data set from Truckads.com. We assign 40 DMAs to the treatment and 166 DMAs to the control group, eliminating four DMAs due to missing observations.

4.3.1. Variables and Descriptive Statistics. Our dependent variable is total number of online purchase transactions (*ONLINE PURCHASES*) in each DMA, aggregated by month. To control for potential economic differences among DMAs, we include various DMA level controls including percent unemployed (*UNEMPLOYMENT*), total number of retail establishments (*RETAIL ESTABLISHMENTS*), and grocery sales (*GROCERY SALES*). As for the offline channel analysis, we include year-month fixed effects to control for seasonality and DMA fixed effects to control for time-invariant characteristics of DMAs. We also include the number of stores (count) and median customer distances for DS-B and DS-C in each DMA. Table 4 reports the descriptive statistics (means and standard deviations) of key variables for the experimental group DMAs before and after the store closures.

As for the online channel, we check the DID parallel trends assumption for the preintervention period

Table 3 High Value vs. Low Value Purchases

Variables	High value				Low value			
	Model-1		Model-2		Model-3		Model-4	
<i>LOW TREAT * CLOSURE</i>	0.03*	(0.012)	0.06***	(0.015)	0.02	(0.012)	0.03	(0.016)
<i>HIGH TREAT * CLOSURE</i>	0.06**	(0.021)	0.09***	(0.022)	0.04	(0.020)	0.05*	(0.022)
<i>CONTROLS</i>	Yes		Yes		Yes		Yes	
<i>MONTH-YEAR FE</i>	Yes		Yes		Yes		Yes	
<i>STORE FE</i>	Yes		Yes		Yes		Yes	
<i>MATCHED SAMPLE</i>	No		Yes		No		Yes	
<i>N</i>	6768		1200		6768		1200	
<i>R²</i>	0.76		0.71		0.89		0.85	

Note: Standard errors (in parentheses) are heteroscedasticity robust and clustered at the store level. FE denotes fixed effects.

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Table 4 Descriptive Statistics—Online Channel

Variables	Treatment group (40 DMAs)				Control group (166 DMAs)			
	Before closures		After closures		Before closures		After closures	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
ONLINE PURCHASES (000)	0.94	1.44	1.20	1.85	0.52	0.84	0.66	1.09
ONLINE SALES (000)	43.97	61.89	58.16	80.17	22.20	34.68	28.73	44.81
UNEMPLOYMENT (%)	4.63	0.98	4.02	0.72	4.81	1.88	4.22	1.58
RETAIL ESTABLISHMENTS	7.21	2.48	5.66	2.43	7.50	2.76	6.96	2.69
GROCERY SALES (000,000)	10.97	12.32	13.38	15.90	3.51	3.94	3.95	4.59
DS-B STORE COUNT	8.95	12.55	8.63	12.38	2.15	3.57	2.15	3.57
DS-C STORE COUNT	9.80	7.07	8.80	7.25	4.02	3.59	4.02	3.59
DS-B MEDIAN DISTANCE (miles)	20.98	25.81	21.30	25.67	71.49	168.19	71.49	168.19
DS-C MEDIAN DISTANCE (miles)	8.26	2.50	9.90	4.03	30.67	189.84	30.67	189.84

before testing H2A. Let $TREAT_d$ denote whether or not a competitor store closure occurs in DMA d . We find that the coefficient for the interaction term, $TREAT_d * TREND_t$, is not statistically significant ($p > 0.1$, $R^2 = 0.56$). With those statistical results, and a visual inspection of the plotted trends (not presented), we find no evidence that the DID parallel trend assumption is violated.

Like our offline channel analysis, we adopt an intensity-based DID approach to assess the impact of distance on experimental units for the online channel and thereby test hypotheses H2A–H2C. Here, however, we categorize DMAs as “high” or “low” treatment intensity based on the weighted median customer distance to the closest DS-A store (as opposed to the distance between a closed store location and the closest DS-A store that we use for the offline channel). To calculate the new distance measure, we identify the corresponding zip codes for each DMA using a data set we collected through Truckads.com. Subsequently, we identify the Haversine distance of each zip code to the closest DS-A store using the longitude and latitude information of zip codes and DS-A store locations. After merging the distance information with zip code population data (acquired through the Unified Data System), we calculate the weighted median customer distance to the closest DS-A store in each DMA. Consistent with Gallino and Moreno (2014), Gallino et al. (2017), and Akturk and Ketzenberg (2021), we use 50 miles as the cutoff threshold for categorizing the high and low intensity treatment groups. Of the 40 DMAs in which DS-B and DS-C close physical stores, 17 are classified as high intensity ($HIGH TREAT_d$) and the balance as low intensity ($LOW TREAT_d$). See section 5 for model robustness with respect to other cutoff distance thresholds. To test H2A, we use the following econometric model:

$$\begin{aligned} \log(ONLINE PURCHASES_{dt}) &= \mu_d + \theta_t + \beta_1 LOW TREAT_d * CLOSURE_t \\ &+ \beta_2 HIGH TREAT_d * CLOSURE_t \\ &+ \beta_m CONTROLS_{m dt} + \varepsilon_{dt} \end{aligned} \quad (2)$$

where μ_d and θ_t are fixed effects for DMAs and month-years, respectively, and ε_{dt} is the error term. Like the offline channel analysis, the vector $CONTROLS$ includes all control variables and the subscript m identifies a specific variable within the vector. We use the FE model for estimation and report heteroscedasticity robust standard errors clustered at the DMA level. Subsequent online channel analyses use the same approach.

We report the results for Equation (2) in Table 5 and use the coefficients for the two-way interaction terms, β_1 and β_2 , shown in rows two and three, to test H2A. The second column (Model-O1) reports the findings with all DMAs included in the analysis. In the third column (Model-O2), we report our findings with propensity score matching such that there are 40 DMAs in each experimental group. See section A.2 for

Table 5 Impact of Store Closures on Online Channel Purchases

Variables	Online purchases			
	Model-O1		Model-O2	
LOW TREAT * CLOSURE	0.01	(0.016)	0.02	(0.018)
HIGH TREAT * CLOSURE	0.09***	(0.022)	0.09***	(0.023)
TIME-VARYING CONTROLS	Yes		Yes	
MONTH-YEAR FE	Yes		Yes	
DMA FE	Yes		Yes	
MATCHED SAMPLE	No		Yes	
N	4944		1920	
R ²	0.63		0.72	

Note: Standard errors (in parentheses) are heteroscedasticity robust and clustered at the DMA level. FE denotes fixed effects.

*** $p < 0.001$.

details of the propensity score matching. The number of observations (N) across the two models differs because the number of DMA s in each model differs. The results for Model-O1 indicate that competitor store closures do not have a statistically significant impact on DS-A's online purchases in low treatment DMA s ($\beta_1 = 0.01$, $p > 0.05$), but have a positive and statistically significant impact on DS-A's online purchases in high treatment DMA s ($\beta_2 = 0.09$, $p < 0.001$). The propensity score matching results for Model-O2 are similar. Since low treatment DMA s are unaffected, H2A is only partially supported.

To test H2B, we investigate the impact of store closures by DS-B and DS-C on online purchases returned to DS-A stores. Defined as RTS instances, products returned to stores during the period of analysis numbered approximately 75K. To test H2B, we use the econometric model in Equation (3):

$$\begin{aligned} \log(RTS_{dt}) = & \mu_d + \theta_t + \beta_1 LOW\ TREAT_d * CLOSURE_t \\ & + \beta_2 HIGH\ TREAT_d * CLOSURE_t \\ & + \log(ONLINE\ PURCHASES_{dt}) \\ & + \beta_m CONTROLS_{mdt} + \varepsilon_{dt} \end{aligned} \quad (3)$$

where RTS_{dt} refers to the number of online purchases returned to DS-A stores in $DMA\ d$ at time t . We include $ONLINE\ PURCHASES_{dt}$ as a control variable for this analysis because the number of returned products depends on the number of online purchases. We report our results in Table 6. Model-C1 (Model-C2) corresponds to the model without (with) propensity score matching. For Model-C1, we find that the coefficient for $LOW\ TREAT_d * CLOSURE_t$ is not statistically significant, whereas the coefficient for $HIGH\ TREAT_d * CLOSURE_t$ is positive and statistically significant ($\beta_2 = 0.06$, $p < 0.05$). The results with propensity score matching (Model-C2) provide a stronger result ($\beta_2 = 0.10$, $p < 0.01$). Because competitor store closures have a

Table 6 Impact of Store Closures on Cross Channel Returns

Variables	Cross channel returns	
	Model-C1	Model-C2
<i>LOW TREAT * CLOSURE</i>	−0.07 (0.061)	−0.03 (0.066)
<i>HIGH TREAT * CLOSURE</i>	0.06* (0.032)	0.10** (0.039)
<i>ONLINE PURCHASES</i>	0.81*** (0.052)	0.86*** (0.097)
<i>TIME-VARYING CONTROLS</i>	Yes	Yes
<i>MONTH-YEAR FE</i>	Yes	Yes
<i>DMA FE</i>	Yes	Yes
<i>MATCHED SAMPLE</i>	No	Yes
<i>N</i>	3205	1317
<i>R</i> ²	0.38	0.38

Note: Standard errors (in parentheses) are heteroscedasticity robust and clustered at the DMA level. FE denotes fixed effects.

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Table 7 Impact of Store Closures on Showrooming Products

Variables	Showrooming	
	Model-S1	Model-S2
<i>LOW TREAT * CLOSURE</i>	0.04 (0.044)	0.06 (0.047)
<i>HIGH TREAT * CLOSURE</i>	0.19** (0.071)	0.18** (0.070)
<i>TIME-VARYING CONTROLS</i>	Yes	Yes
<i>MONTH-YEAR FE</i>	Yes	Yes
<i>DMA FE</i>	Yes	Yes
<i>MATCHED SAMPLE</i>	No	Yes
<i>N</i>	4721	1897
<i>R</i> ²	0.40	0.42

Note: Standard errors (in parentheses) are heteroscedasticity robust and clustered at the DMA level. FE denotes fixed effects.

** $p < 0.01$.

statistically measurable effect on RTS instances in high treatment but not in low treatment DMA s, H2B is only partially supported.

We test H2C with the apparel product category (310K transactions), a top showrooming category (Khan 2018, Richter 2013). We report our results in Table 7. Model-S1 (Model-S2) corresponds to the model without (with) propensity score matching. For Model-S1, we find that the coefficient for $LOW\ TREAT_d * CLOSURE_t$ is not statistically significant, whereas the coefficient for $HIGH\ TREAT_d * CLOSURE_t$ is positive and statistically significant ($\beta_2 = 0.19$, $p < 0.01$). The results for propensity score matching in Model-S2 are similar ($\beta_2 = 0.18$, $p < 0.01$). Because competitor store closures have a statistically measurable effect on showrooming category products in high treatment but not in low treatment DMA s, H2C is only partially supported.

To present a complete picture, we also examine the effect that store closures have on nonshowrooming category products. To do so, we replicate the same analysis just presented for the showrooming category products but now using the transactional record of products which are either not suitable for showrooming or for which customers exhibit limited showrooming behavior. These include products available only via DS-A's online channel (265K transactions) as well as jewelry products (47K transactions). Note that industry sources report low showrooming activity for jewelry (e.g., Khan 2018, Richter 2013). We report our results in Table 8. Model-NS1 (Model-NS2) corresponds to the model without (with) propensity score matching. As the coefficients for the two-way interaction terms are not statistically significant in either model, the results in Table 8 provide additional support for H2C. In high treatment DMA s, we see an increase in sales for showrooming product categories, but not for product categories in which showrooming is largely irrelevant.

Table 8 Impact of Store Closures on Non-Showrooming Products

Variables	Non-showrooming			
	Model-NS1		Model-NS2	
<i>LOW TREAT</i> * <i>CLOSURE</i>	0.02	(0.027)	0.01	(0.037)
<i>HIGH TREAT</i> * <i>CLOSURE</i>	0.11	(0.064)	0.09	(0.067)
<i>TIME-VARYING CONTROLS</i>	Yes		Yes	
<i>MONTH-YEAR FE</i>	Yes		Yes	
<i>DMA FE</i>	Yes		Yes	
<i>MATCHED SAMPLE</i>	No		Yes	
<i>N</i>	3984		1632	
<i>R</i> ²	0.61		0.65	

Note: Standard errors (in parentheses) are heteroscedasticity robust and clustered at the *DMA* level. FE denotes fixed effects.

5. Robustness Checks

Further support for our findings is provided by robustness checks that include tests (i) to rule out concerns that the increased sales observed for both channels do not originate from existing customers, (ii) regarding the level of aggregation, (iii) for internal validity (i.e., placebo effects), (iv) for distance cutoff thresholds, (v) concerning cross channel return rates, (vi) to rule out concerns regarding the impact of DS-A's own store openings and closures, (vii) to address potential concerns regarding price discounts, and (viii) concerning the potential heterogeneous impact of competitor store closures.

To explore whether the increase in DS-A's offline channel sales arise from existing or newly acquired customers, we first identify customers who shopped at DS-A in both the pre-closure and post-closure periods, denoting these customers as "existing customers." We identify 51.2 million transactions completed by existing customers during the period of analysis and replicate our original analysis for the offline channel using these customers. We find that the increase in purchases observed for the treatment group DS-A stores in our original analysis (see Table 2) does not arise from existing customers. Conducting the same analysis for the online channel, we identify 2.25 million transactions completed by existing customers and find that the increase in purchases observed for the treatment group *DMA*s in our original analysis (see Table 5) is not attributable to existing customers.

Although each transaction in our data set is date and time stamped, we perform our primary analyses at the monthly aggregation level. More time periods are available with weekly aggregation, but we lose 25 *DMA*s due to missing observations for the dependent variable. We nevertheless replicate our analysis using weekly aggregation as a test of model robustness. We have 104 weeks for this analysis, 52 weeks before and

52 weeks after. The weekly level results are consistent with the monthly level results reported in Table 5. Performing the same analysis for the offline channel yields results consistent with those reported in Table 2.

We also perform tests to address potential internal validity concerns associated with using the DID methodology. To check whether the structure of the data is solely responsible for our findings (Bertrand et al. 2004), we perform placebo tests in the pre-closure period to see if choosing a random (and artificial) month for store closures yields the same results. Conducting our analyses at both monthly and weekly levels of aggregation, the variables of interest are not statistically significant in any model.

Our DID analysis is predicated on specific distance measures for both offline and online channels. For the offline channel, we use 20 miles (measured as the distance between a closed store location and the closest DS-A store) as the distance threshold for assigning stores to treatment and control groups. Replicating the analyses at different distances (10, 15, and 30 miles) for robustness does not change our results. Nor does replicating the analysis using a 10-mile threshold to distinguish stores assigned to high and low treatment groups with distance thresholds of 5 and 7 miles change the results. As for the online channel, we use a 50-mile distance threshold (weighted median customer distance to the nearest DS-A store within a *DMA*) to distinguish low and high treatment *DMA*s. Replicating this analysis with distance thresholds of 30, 40, and 60 miles does not change the results.

Although we use the number of RTS instances in our analysis of cross channel returns, we also use "return rate" as the response variable by calculating the ratio of RTS instances to the total number of online purchase transactions using the model by Papke and Wooldridge (2008), which can handle fractional response variables in a longitudinal setting through a probit link function. We find that the coefficient for $HIGH\ TREAT_d * CLOSURE_t$ is positive and statistically significant ($\beta_2 = 0.03$, $p < 0.01$) and the coefficient for $LOW\ TREAT_d * CLOSURE_t$ not statistically significant. Hence, our findings are consistent with the results reported in Table 6.

We also perform additional tests to rule out concerns that DS-A's own store openings and closures might confound our findings. For the offline channel, we find that 12 stores may have been impacted by DS-A's own store closures and openings. Hence, we drop these 12 DS-A stores from the analysis and use the remaining 270 stores. Likewise for the online channel analysis, finding that seven *DMA*s may have been affected by DS-A's own store openings and closures, we drop these *DMA*s from the analysis and use the

remaining 199. Even so, the results are consistent with those we report in Tables 2 and 5.

Furthermore, to rule out potential concerns that price discounts may be confounding our results, we present visual evidence by plotting monthly average price of purchased items for the treatment and control groups across both offline and online channels. We present the results for the offline channel in Online Appendix A.5 and for the online channel in Online Appendix A.6. Plots for both channels reveal average monthly prices follow a similar pattern for both experimental groups.

Lastly, we test for heterogeneity effects across the online and offline channels. We present the results for the offline channel in Online Appendix A.7 and for the online channel in Online Appendix A.8. We find that DS-A's offline channel sales increase more for DS-A stores for which competitors' proximity is "low" relative to "high" and for DS-A stores facing "weak" (competitor stores are far away) relative to "strong" (competitor stores are nearby) competition. For the online channel, we find that DS-A's online sales increase more in DMAs in which competitors have "low" relative to "high" proximity to customers.

6. Managerial Insights and Conclusion

Our research, to the best of our knowledge, is the first to empirically investigate the competitive impact of store closures on a major retailer. That a store closure forces former patrons to seek out a new shopping alternative, and that some are likely to choose a nearby competitor, is not surprising in today's hyper-competitive retail market. To date, however, such customer defections have not been directly assessed using the transactional record of a competitor, nor the effect of distance empirically investigated. More interestingly, however, we find that competitor store closures impact both offline and online channels of the focal retailer.

We analyze each channel separately. For the offline channel, sales increase by an average of 2.8% for stores in the treatment group and distance to the closed store locations affects the intensity of the sales increase. After store closures, sales increase by an estimated 2.2% for DS-A stores located between 10 and 20 miles, translating to an average increase of \$64K and ranging between \$33K and \$96K. Sales at stores located within 10 miles increase by an estimated 3.6%, corresponding to an average of \$89K and ranging between \$40K and \$169K.

For the online channel, we hypothesized that online sales would increase in geographic markets in which competitors close stores, and that the increase would be greater for regions in which DS-A stores are located in close proximity to customers. This

hypothesis is only partially supported as we find that sales in low intensity treatment DMAs do not increase. We estimate sales in high intensity treatment DMAs to increase by an estimated 9.3%, corresponding to an average of \$82K and ranging between \$11K and \$369K. While we expected the impact to increase with proximity, our robustness tests that use increasingly smaller cutoff thresholds are not supporting. Perhaps aggregation at the DMA level is not able to sufficiently discern treatment intensity. Overall, we estimate that online sales increase by \$1.4M and offline sales by \$1.9M after the competitor store closures. Note that of the total estimated increase, 42.6% is from the online channel, which as a whole, accounts for only 8.3% of total sales.

Observing that sales in both channels increase, we shift our attention to the underlying mechanisms that explain the increase. To gain insight into these mechanisms, we developed a set of hypotheses predicated on the value proposition stores offer customers and the expected behavioral responses of those customers when faced with a store closure. A store closure forces the choice of an alternative, and it is on the basis of the tradeoff between substitution costs and transactions costs that the choice is made. All else being equal, closer alternatives are better, and this reasoning falls in line with our DID intensity-based results and the underlying transaction cost theory that explains them. For offline shoppers, stores need to be nearby. But our results are also consistent with the value online shoppers also place on nearby stores. Many consumers prefer to shop online and then make their purchases in stores, and this is consistent with our observation that sales at DS-A stores increase more for high value than for low value items, even in the presence of a preponderance of low value transactions. High value items, being riskier, tend to invite webrooming, whereby consumers mitigate uncertainty by researching their purchases online prior to purchasing them in stores (Flavián et al. 2016).

Stores provide a greater level of delivery and shopping assurance than is generally available online. Delivery assurance comes from fast, free, and secure order fulfillment, while shopping assurance arises from the ability to reduce purchase uncertainty and associated risks through the touch and feel experience and the RTS omnichannel service. In geographic regions associated with a sales increase, the increase is observed for product categories that lend themselves to showrooming, but not in categories for which showrooming is immaterial or unpopular. Moreover, we observe an increase in RTS instances in affected regions in which DSA-A has stores located in close proximity to customers. Hence, stores are important to customers whether they decide to shop online and make purchases in stores or shop in stores

and make purchases online. In either case, it is apparent that a nearby store location is integral to the ultimate purchase decision for many consumers.

The results for both channels have significant implications for strategies that concern downsizing and closing store locations. Competitive delocation models that optimize store closure decisions generally treat demand as exogenous and assume that the closest remaining open store, whether the same brand or a rival, will capture all former demand of a closed store, and that the demand will be split in the case of ties. The underlying assumption is that a store, located within a threshold distance customers are willing to travel to obtain a good, serves all demand equally, irrespective of distance (e.g., Mayadunne et al. 2018, ReVelle et al. 2007). This assumption, however, ignores the direct effect of distance that decreases the size and frequency of customer visits due to the associated transaction costs. As distance increases, demand is likely to shift to substitute goods and channels. Moreover, none of the analytical models of which we are aware address the impact store closures have on the online channel, as our empirical results demonstrate.

Overall, the effects of store closures highlight the many and varied roles that stores play in retailing. As such, it is meaningful and relevant to reassess and reimagine the operational design choices with respect to them. Consider webrooming and showrooming that play a significant role in customer search. When stores close, these shopping habits are disrupted, requiring customers to seek an alternative. Instead of closing stores, retailers might consider downsizing to small showrooms. A case in point, Nordstrom is investing in neighborhood stores throughout the United States that effectively serve as showrooms and stock only the most popular items. Whereas a typical Nordstrom location is 140,000 square feet, a neighborhood store is about 3,000 square feet (Thomas 2017). These local stores are also intended to boost online sales through omnichannel services and express alterations (Nordstrom 2020). That said, competitors seeking to take advantage of rival store closures, but without a nearby store, might consider doing the same. Opening small, conveniently located showrooms can be an effective way to gain a foothold in new markets and to capture demand from competitors that downsize.

As for webrooming, stores provide shopping assurance, motivating customers to purchases in stores after shopping online. A retailer in a given market that closes stores and no longer has nearby locations should endeavor to provide that shopping assurance via other means like free returns and web technologies like zoom, alternative photos, and color swatches (De et al. 2013). Enabling returns at other retailers'

stores, such as the partnership that allows Amazon customers to make returns at Kohls' stores, is another option. Another unique Amazon service that increases shopping assurance is Prime Wardrobe, which encourages customers to try products prior to purchasing them. The key is for retailers to find new ways to provide services for online customers who are lost when stores are closed.

Naturally, there are plenty of opportunities for future research, and considerably more is needed to understand the effects of store closures, how to optimally make store closure decisions, and determine best competitor responses to such decisions. Although we examine the transactional record to gain insight into customer behavior, we can only indirectly assess customer motivations and preferences. Future research could triangulate using other research methods like laboratory experiments and customer surveys to gain a better understanding of the customer decision-making process and key determinants of patronage when stores are closed. Furthermore, our analysis occurs wholly through the lens of transaction costs and ignores substitution costs. A better understanding of the tradeoff between the two will help to inform the design and implementation of decision support models. How do loyalty intentions of customers change after store closures? Answering these and similar questions will help improve retail decision-making.

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Supporting Information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Appendix A: Online Appendix.